

2023-Ph-H1-Q16

Section: Particles and Waves

Topic: Wave-Particle Duality

Which experiment provided evidence for wave-particle duality of electrons?

Answer: D

(Electron diffraction)

Guidance for Students:

Electrons, though particles, can exhibit wave-like behavior. When passed through a crystal, they diffract — just like light waves — proving they have wave properties.

Revision Tips:

- Electron diffraction shows interference patterns.
- Wave-particle duality: matter has both wave and particle properties.
- Key equation: $\lambda = \frac{h}{p}$ (de Broglie)